From the Chain Rule to ChatGPT: The Mathematics Behind AI

WAMATYC 2026 — Conference Handouts

John Mitchell · Clark College

jmitchell@clark.edu

Contents

This bundle contains seven handouts to accompany my talk and slides. All files are provided as PDF.

File	Title
`WAMATYC_AI_Handout1.pdf`	1. Regression and the Single Neuron
`WAMATYC_AI_Handout2.pdf`	2. Feedforward Networks and Backpropagation
`WAMATYC_AI_Handout3.pdf`	3. The Transformer and Large Language Models
`WAMATYC_AI_Handout4.pdf`	4. Connections to Calculus
`WAMATYC_AI_Handout5.pdf`	5. Connections to Statistics
`WAMATYC_AI_Handout6.pdf`	6. Connections to Linear Algebra
`WAMATYC_AI_Handout7.pdf`	7. Glossary of AI and Machine Learning Terms

How to Use These Handouts

Handouts 1–3 follow the narrative arc of the talk. They are intended as reference documents for a deeper understanding of the ideas covered in the slides.

Handouts 4–6 are discipline-specific bridge documents for instructors who teach calculus, statistics, or linear algebra. Each maps the AI concepts from the talk to specific topics in the relevant course sequence, and includes a sample activity for classroom use.

Handout 7 is a glossary of AI and machine learning terminology, intended as a reference for attendees unfamiliar with the field. It covers hardware, core concepts, neural network architecture, language model terminology, and training methods.

Source and Attribution

These handouts were developed collaboratively with Claude Sonnet 4.6 (Anthropic) through an iterative conversation. Source files are written in Markdown with LaTeX equations and converted to PDF using Pandoc and XeLaTeX. Source files are available on request.